04-train-validate-test

April 21, 2026

1 CSE 25 - Introduction to Artificial Intelligence

1.1 Week 4 Tuesday: Using a model to predict and generalize

We used gradient descent to optimize parameters in a linear model for *regression* tasks and we used the perceptron update role to optimize parameters in a linear model for *classification*. Our focus was on *fitting* the model to the data we were given.

Today's guiding questions:

- What can we do *after* we fit a model?
- How do we know if a model works well on new data?

Learning Objectives:

By the end of today's class, you will be able to:

- Describe why the perceptron update rule works for binary settings.
- Define perceptron loss.
- Explain the purpose of **train / validation / test** splits
- Define and describe **generalization**, **overfitting**, and **underfitting**
- Interpret a **training** curve
- Interpret a **validation** curve

Instructions

Use your copy of this notebook on Datahub and complete it during class. Work through the cells below **in order**. You may discuss with your neighbors, but make sure you understand each step yourself.

SUBMISSION: When finished, download it as a PDF and upload it to Gradescope under `In-Class - Week 4 Tuesday` to receive credit. To download it as a PDF, on DataHub go to `File -> Save and Export Notebook As -> PDF`.

1.1.1 From last time: How does the perceptron learn?

The perceptron is a linear model for classification and learns by updating its weights and bias whenever it makes a mistake on a training example.

- For each training example, it computes the score:

$$score = z = \sum_{i=1}^{n} w_i x_i + b$$

- It predicts the class based on the sign of the score.
  - If the score is greater than or equal to 0, predict one class (+1)
  - If the score is less than 0, predict the other class (-1)
- If the prediction is wrong, it *updates* the weights and bias to reduce future mistakes:

  Update rule:

  - For each feature i:
    * $w_i = w_i + y \cdot x_i$
  - Bias:
    * $b = b + y$

  Here, y is the true label (+1 or -1), and x_i is the value of feature i.

This update nudges the model to be more likely to predict the correct class next time for similar examples.

An **epoch** is one iteration of this process through the training data.

1.1.2 Perceptron Loss

The perceptron update can be understood as reducing the **perceptron loss**.

The perceptron loss measures how badly a point is misclassified by the model. The perceptron loss is given by:

$$\mathcal{L}(y, z) = \max(0, -yz)$$

where z is the score for the current data point and y is the true label which is in $\{-1, 1\}$.

This loss is designed to be small when the model classifies a point correctly and large when it classifies a point incorrectly.

Why does this make sense?

- If $yz > 0$, then $-yz < 0$.
  - In this case the loss is 0.
  - Notice that $yz > 0$ only happens when y and z have the same sign.
  - There are two possible cases.
    * *Case 1*: y and z are both positive, and we have a TP (the model predicts positive because $z > 0$ and the true label is positive because $y > 0$ so $y = 1$). `This is good model behavior!`
    * *Case 2*: y and z are both negative, and we have a TN (the model predicts negative because $z < 0$ and the true label is negative because $y < 0$ so $y = -1$). `This is good model behavior!`
- If $yz < 0$ then $-yz > 0$.
  - In this case the loss is $-yz$ and is positive.
  - Notice that $yz < 0$ only happens when y and z have different signs.
  - There are two possible cases.

* *Case 1*: z is positive and y is negative, so we have a FP (the model predicts positive because $z > 0$ and the true label is negative because $y < 0$ so $y = -1$) `This is bad model behavior!`
* *Case 2*: z is negative and y is positive, so we have a FN (the model predicts negative because $z < 0$ and the true label is positive because $y > 0$ so $y = -1$) `This is bad model behavior!`

`The loss is zero for clearly correct predictions and positive for (most) incorrect predictions.`

What about the boundary case, $z = 0$? If $z = 0$ then the point is exactly on the decision boundary. In this case, $yz = 0$ so loss is 0. But, y is either 1 or -1. Since the model predict label 1 in this case, the perceptron loss does not perfectly capture this boundary case. Regardless, the perceptron training algorithm will update the model in this case because y won't match the predicted label.

Why the Update Reduces Loss For a misclassified point, $yz \leq 0$. When the model weights are updated:

- For each feature i, w_i is updated to be $w_i + yx_i$.
  - If $y = 1$, this changes z in the direction of x_i
  - If $y = -1$, this changes z in the opposite direction of x_i
- The bias is updated up if $y = 1$ and down if $y = -1$.

Using some linear algebra (stay tuned for Math 18), the new score after the update will be

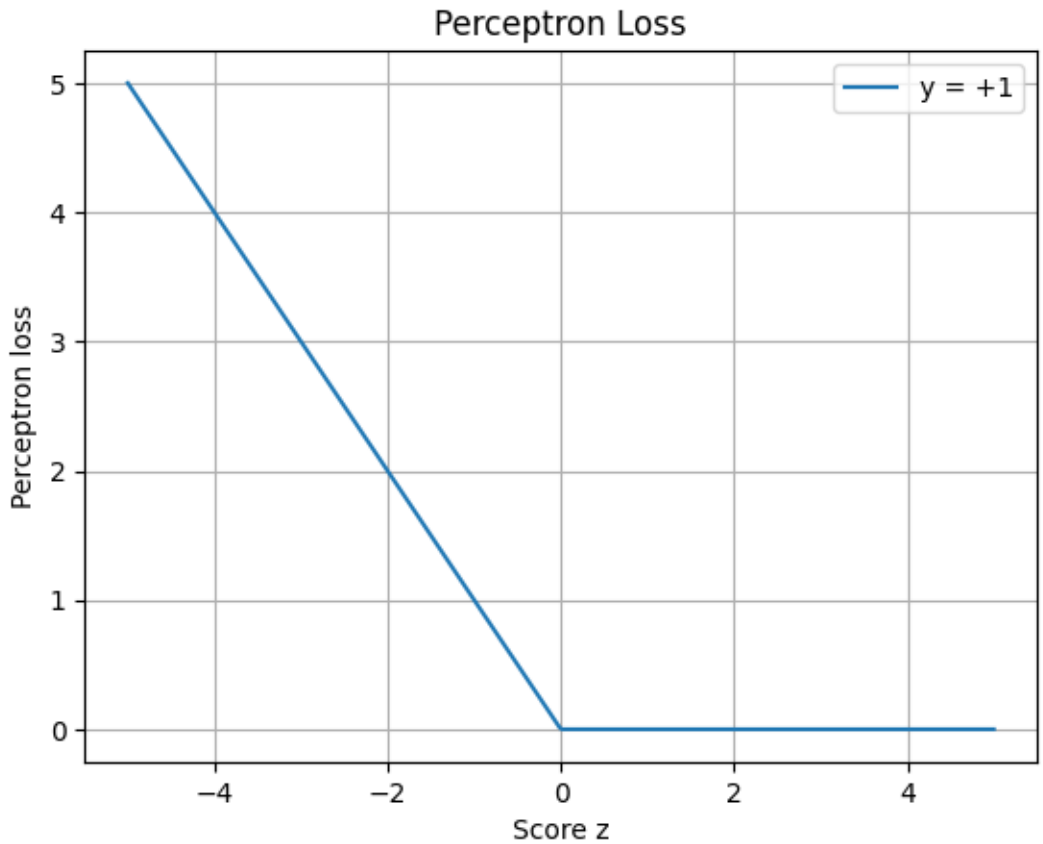
$$\text{new score} = \text{old score} + y(\text{a positive number related to all the } x_i)$$

- The new score is `less than` the old score if $y = -1$: more likely to be less than 0 and lead to the model classifying this point as -1. - The new score is `greater than` the old score if $y = +1$: more likely to be greater than or equal to 0 and lead to the model classifying this point as 1.

We can visualize the relationship between perceptron score and perceptron loss.

[1]:
```
import numpy as np
# Import the matplotlib library for plotting
import matplotlib.pyplot as plt

# Range of scores
z = np.linspace(-5, 5, 400)

# Perceptron loss
loss_y_pos = np.maximum(0, -z)   # y = +1
loss_y_neg = np.maximum(0, z)    # y = -1

plt.figure()
plt.plot(z, loss_y_pos, label="y = +1")

# Uncomment the next line to see the loss function for when y = -1
#plt.plot(z, loss_y_neg, label="y = -1")

plt.xlabel("Score z")
```

```
plt.ylabel("Perceptron loss")
plt.title("Perceptron Loss")
plt.grid(True)
plt.legend()
plt.show()
```



[2]:
```
import matplotlib.pyplot as plt
from ipywidgets import interact, FloatSlider
import numpy as np  # Used only for fast grid rendering
import math

# Data using standard Python lists (labels in {-1, +1})
X_toy = [[1.5, 4.0], [1.0, 2.0], [2.0, 1.0], [3.0, 5.0], [4.0, 2.0], [0.0, 0.
↪0], [1.5, -0.5]]
y_toy = [1, 1, -1, 1, -1, -1, -1]

def combined_perceptron_plot(w1=1.0, w2=1.0, b=-2.0):
    '''
    Display two views of data set and perceptron model performance.
```

```
    w1,w2: model weights (default: 1.0)
    b: model bias (default: -2.0)
    '''
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

    # LEFT: 1D Step View (sign)
    z_axis = [z * 0.1 for z in range(-100, 101)]
    ax1.plot(z_axis, [1 if z >= 0 else -1 for z in z_axis],
             color='green', drawstyle='steps-mid', label="sign(z)")
    ax1.axvline(x=0, color='black', linestyle='--', alpha=0.5)

    total_loss = 0  # Initialize loss counter

    for i in range(len(X_toy)):
        x1, x2 = X_toy[i]
        y = y_toy[i]
        z_score = w1 * x1 + w2 * x2 + b
        pred = 1 if z_score >= 0 else -1

        loss = max(0.0, -y * z_score)  # Perceptron loss
        total_loss += loss

        color, m = ('blue', 'x') if y == 1 else ('red', 'o')
        if y == 1:
            ax1.scatter(z_score, pred, color=color, marker=m, s=100,␣
↪linewidths=2, zorder=5)
        else:
            ax1.scatter(z_score, pred, color=color, marker=m, s=80,␣
↪edgecolors='k', zorder=5)

    # Update title with the calculated loss
    ax1.set_title(f"Perceptron: Score vs Prediction\nTotal Perceptron Loss:␣
↪{total_loss:.2f}")
    ax1.set_xlabel("Score (z)")
    ax1.set_ylim(-1.1, 1.1)
    ax1.set_yticks([-1, 1])

    # RIGHT: 2D Feature Space (Speedy Rendering)
    xx, yy = np.meshgrid(np.linspace(-1, 5, 100), np.linspace(-1, 6, 100))
    zz = w1 * xx + w2 * yy + b

    # Binary background: everything <0 is blue, everything >0 is red
    ax2.contourf(xx, yy, zz, levels=[-1e9, 0, 1e9], colors=['red','blue'],␣
↪alpha=0.2)

    if w2 != 0:
```

```
        x_line = np.array([-1, 5])
        y_line = -(w1 * x_line + b) / w2
        ax2.plot(x_line, y_line, "k-", linewidth=2)

    for i in range(len(X_toy)):
        x1, x2 = X_toy[i]
        y = y_toy[i]
        if y == 1:
            ax2.scatter(x1, x2, color='blue', marker='x', s=100, linewidths=2)
        else:
            ax2.scatter(x1, x2, color='red', marker='o', s=80, edgecolors='k')
        # label each point with its coordinates
        ax2.annotate(
        f'({x1}, {x2})',          # text to display
        xy=(x1, x2),              # point the text refers to
        xytext=(5, 5),            # offset in points from the marker (right and␣
↪up)
        textcoords='offset points',
        fontsize=10,
        color='black',
        )

    ax2.set_xlim(-1, 5)
    ax2.set_ylim(-1, 6)
    ax2.set_title("2D View: Hard Decision Regions")
    plt.tight_layout()
    plt.show()

interact(combined_perceptron_plot,
         w1=(-5.0, 5.0, 0.1),
         w2=(-5.0, 5.0, 0.1),
         b=(-10.0, 10.0, 0.1))
```

```
interactive(children=(FloatSlider(value=1.0, description='w1', max=5.0, min=-5.
↪0), FloatSlider(value=1.0, desc…
```

[2]:
```
<function __main__.combined_perceptron_plot(w1=1.0, w2=1.0, b=-2.0)>
```

Q: Match the points on the left graph with the points on the right graph

YOUR ANSWER HERE

Q: For the default model parameters, what is the Total Perceptron Loss, as displayed on the graph?

YOUR ANSWER HERE

Q: Calculate this number directly from the definition:

$$\text{Total perceptron loss} = \sum_{data} -y \left(\sum_{i=1}^{n} w_i x_i + b \right)$$

YOUR ANSWER HERE

1.1.3 Graphing training performance

For a binary prediction model that predicts labels +1 (positive) or -1 (negative), the confusion counts for a data set are

- **TP**: number of true positive (true 1, predicted 1) examples
- **FP**: number of false positive (true -1, predicted 1) examples
- **FN**: number of false negative (true 1, predicted -1) examples
- **TN**: number of true negative (true -1, predicted -1) examples

)

`Accuracy`: Of all the examples the model predicted, how many predictions were correct?

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

`Precision`: Of all the examples the model *predicted* as positive, how many were actually positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

`Recall`: Of all the examples that are *actually* positive, how many did the model correctly identify? Recall is also known as *True Positive Rate (TPR)*.

$$\text{Recall} = \frac{TP}{TP + FN}$$

To see how training impacts these model metrics, we'll graph the performance of a model as the model parameter changes during training.

[3]:
```
import random

# Perceptron learning implementation from last time, somewhat reorganized

def perceptron_predict_all(X_list, w, b):
    '''
    Return a list of predictions (+1 or -1) for every point in X_list.
    '''
    return [1 if w[0]*x[0] + w[1]*x[1] + b >= 0 else -1 for x in X_list]

def compute_metrics(y_true, y_pred):
    '''
    Compute accuracy, precision, and recall for binary labels in {-1, +1}.

    Treats +1 as the positive class.

    Returns a tuple (accuracy, precision, recall).
    Precision and recall are set to 0.0 when the denominator would be zero
```

```
    (i.e. the model never predicts positive, or there are no true positives).
    '''
    tp = sum(1 for t, p in zip(y_true, y_pred) if t ==  1 and p ==  1)
    fp = sum(1 for t, p in zip(y_true, y_pred) if t == -1 and p ==  1)
    fn = sum(1 for t, p in zip(y_true, y_pred) if t ==  1 and p == -1)

    accuracy  = sum(1 for t, p in zip(y_true, y_pred) if t == p) / len(y_true)
    precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
    recall    = tp / (tp + fn) if (tp + fn) > 0 else 0.0

    return accuracy, precision, recall

def perceptron_train_with_history(X, y, num_epochs=10):
    '''
    Train a perceptron and record loss, accuracy, precision, and recall
    after every individual data point is processed (across all epochs).

    X: list of [x1, x2] feature pairs
    y: list of labels in {-1, +1}
    num_epochs: number of full passes over the data

    Returns
    w, b : final weights and bias
    history : dictionary with keys 'loss', 'accuracy', 'precision', 'recall',
              each a list of length (num_epochs * len(X))
    '''
    w = [0.0, 0.0]
    b = 0.0

    history = {'loss': [], 'accuracy': [], 'precision': [], 'recall': []}

    rng = random.Random(42)

    for epoch in range(num_epochs):
        # Shuffle a new order each epoch
        combined = list(zip(X, y))
        rng.shuffle(combined)

        for x, y_true in combined:
            # Calculate perceptron score and apply update rule
            score  = w[0] * x[0] + w[1] * x[1] + b
            y_pred = 1 if score >= 0 else -1

            if y_pred != y_true:
                w[0] += y_true * x[0]
                w[1] += y_true * x[1]
                b    += y_true
```

```
            # Calculate current model predictions for all training data
            all_preds = perceptron_predict_all(X, w, b)

            # Total perceptron loss = sum of max(0, -y*z) across all points
            total_loss = sum(
                max(0.0, -yt * (w[0]*xt[0] + w[1]*xt[1] + b))
                for xt, yt in zip(X, y)
            )

            acc, prec, rec = compute_metrics(y, all_preds)

            history['loss'].append(total_loss)
            history['accuracy'].append(acc)
            history['precision'].append(prec)
            history['recall'].append(rec)

    return w, b, history
```

We will plot each of these metrics during a training run. Before we do, let's make some predictions.

Q: At what value do we expect each metric to settle?

YOUR ANSWER HERE

Q: Does the above answer depend on the observation that the toy data set is **linearly separable**?

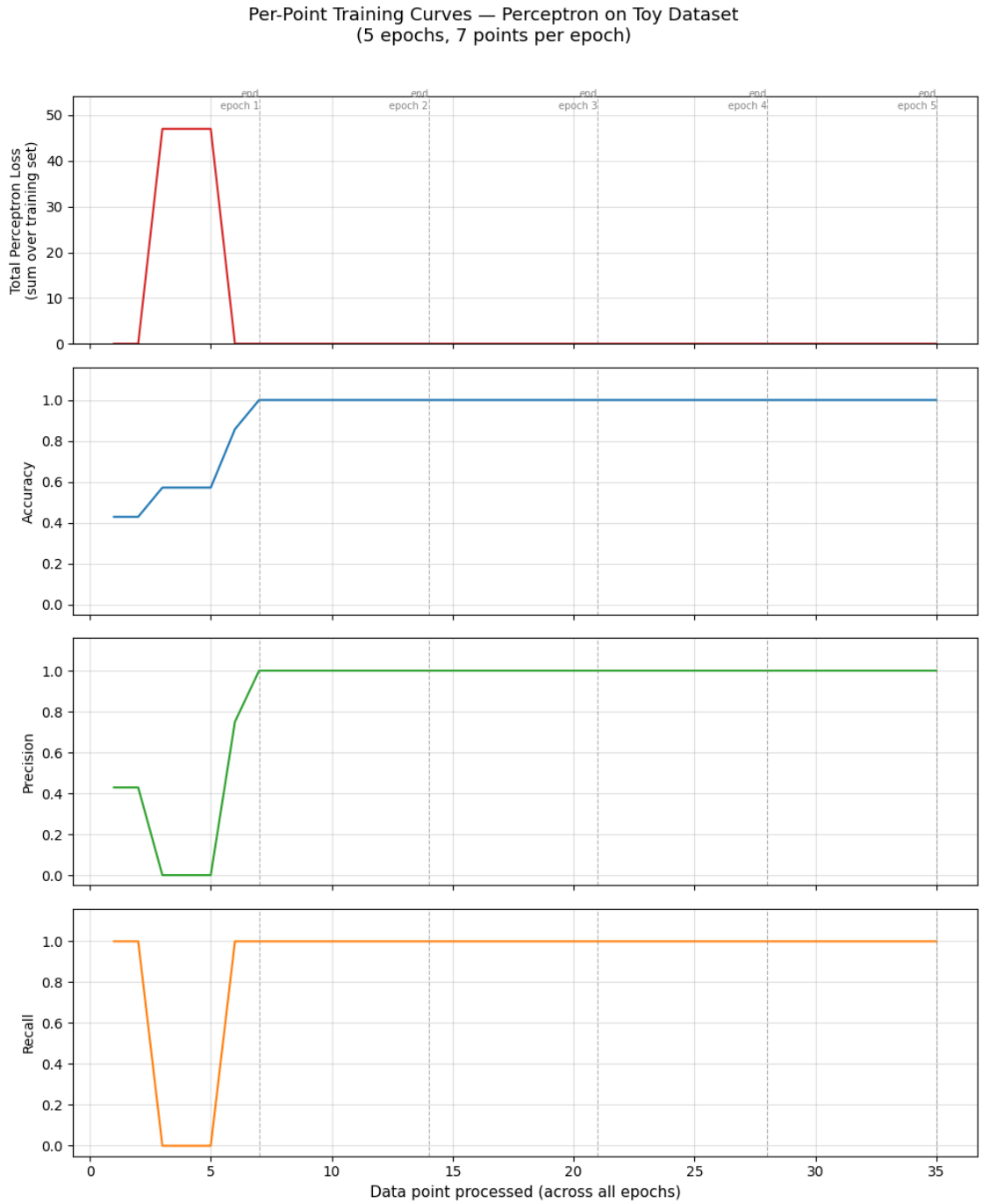
YOUR ANSWER HERE

Q: Do we expect each metric to rise, drop, or oscillate?

[4]:
```
# Plot four training curves with different y axes: perceptron loss and accuracy␣
↪and precision and recall after each point.

NUM_EPOCHS = 5

w_final, b_final, history = perceptron_train_with_history(
    X_toy, y_toy, num_epochs=NUM_EPOCHS
)

n_points   = len(X_toy)
total_steps = len(history['loss'])   # = NUM_EPOCHS * n_points
num_epochs_run = len(history['loss'])

print("Num points in training set is", n_points)
print("Num steps in training process is", total_steps)
print("Num epochs run is ", num_epochs_run)

# x-axis: data-point index (step 1 = after first point of epoch 1, etc.)
steps = list(range(1, total_steps + 1))
```

```python
# Vertical lines mark the end of each epoch
epoch_boundaries = [n_points * e for e in range(1, NUM_EPOCHS + 1)]

#   Four-panel figure
metrics = [
    ('loss',      'Total Perceptron Loss\n(sum over training set)', 'tab:red'),
    ('accuracy',  'Accuracy',                                       'tab:blue'),
    ('precision', 'Precision',                                      'tab:green'),
    ('recall',    'Recall',                                         'tab:orange'),
]

fig, axes = plt.subplots(4, 1, figsize=(10, 12), sharex=True)

for ax, (key, ylabel, color) in zip(axes, metrics):
    ax.plot(steps, history[key], color=color, linewidth=1.5)
    for eb in epoch_boundaries:
        ax.axvline(x=eb, color='gray', linestyle='--', linewidth=0.8, alpha=0.6)
    ax.set_ylabel(ylabel, fontsize=10)
    ax.set_ylim(-0.05, max(history[key]) * 1.15 + 0.01)
    ax.grid(True, alpha=0.4)

# Label epoch boundaries on the top panel only
for i, eb in enumerate(epoch_boundaries):
    axes[0].text(eb, axes[0].get_ylim()[1] * 0.95,
                 f'end\nepoch {i+1}', ha='right', fontsize=7, color='gray')

axes[-1].set_xlabel('Data point processed (across all epochs)', fontsize=11)
fig.suptitle(
    f'Per-Point Training Curves - Perceptron on Toy Dataset\n'
    f'({NUM_EPOCHS} epochs, {n_points} points per epoch)',
    fontsize=13, y=1.01
)
plt.tight_layout()
plt.show()

print(f"Final weights: w = {[round(wi, 3) for wi in w_final]}, b =␣
 ↪{round(b_final, 3)}")
```

```
Num points in training set is 7
Num steps in training process is 35
Num epochs run is  35
```




```
Final weights: w = [-2.5, 2.0], b = -1.0
```

Some observations about the curves:

- **The model converges in a single epoch**
  - All four curves reach their final values by the end of the first epoch.
  - The toy dataset is linearly separable.

  - Once the perceptron finds a correct decision boundary, it correctly classifies every point and the update rule (`if y_pred != y_true: update`) never triggers again. Additional epochs do nothing.
- **The loss curve is not monotonically decreasing**
  - Initial weights `w = [0, 0], b = 0` give score `z = 0` for every point. Since `z  0` predicts $+1$, and 4 of the 7 points are class -1, the first few updates correct some errors but in doing so move the boundary — which may simultaneously *increase* the loss on other points before the boundary settles in the right place.
  - A single update that fixes one mistake may create temporary errors on other points.
  - This is a key difference between the perceptron update rule (that is applied based on one training data example at a time) and gradient-based methods (that average over the whole dataset before updating).
- **Precision and recall start high, then crash low, then recover**
  - The initial model parameters `w = [0, 0], b = 0` mean that the model predicts $+1$ for *every* point.
  - Precision is low because this model predicts that seven examples are positive, but only three actually are.
  - Recall is high because all three of the positive examples are predicted to be positive (and there are no false negatives).
  - The crash in both precision and recall happens when the model briefly predicts everything as -1. In this case, both metrics are 0.
  - When the model converges and finds a good decision boundary, all predictions are true so both precision and recall are 1.

`Accuracy`: Of all the examples the model predicted, how many predictions were correct?

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

`Precision`: Of all the examples the model *predicted* as positive, how many were actually positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

`Recall`: Of all the examples that are *actually* positive, how many did the model correctly identify? Recall is also known as *True Positive Rate (TPR)*.

$$\text{Recall} = \frac{TP}{TP + FN}$$

1.1.4 Train / Validation / Test Split

We've seen how to train a model and how to use a model; how do we know if our model is correct?

It's not enough to just fit a line or decision boundary and make predictions or classify - we want to know if our model is actually making good predictions on unseen data.

When we measure model *accuracy*, *precision*, and *recall* on training data, the results can be misleading. A model might memorize training examples without learning anything general.

In practice, we split our data into two parts:

- **Training data:** Used to fit (train) the model.
- **Evaluation data:** Used to evaluate how well the model predicts on new, unseen data.

This way, we can check if our model *generalizes* well, not just *memorizes* the training points. This means that we train a model on one portion of the data and then evaluate it on *new, unseen examples* to ask:

*Does the model **generalize**, or did it just memorize the training data?*

Evaluation is important because - Training performance is often **better** than test performance. - A model that fits the training data well may still struggle on new data. - Evaluation on unseen data helps us understand whether the model truly learned patterns.

Evaluation happens at several points: **validation** during development of the model, and **testing** at the very end before deployment.

Training set Used to fit model parameters (weights and biases). The model sees these examples during learning.

Validation set Used during development to monitor performance, compare models, tune *hyperparameters*, and detect **overfitting** (when a model learns the training data too closely and performs worse on new data). The model does **not** update its parameters on this data.

Test set Used only once at the very end to report final performance. It acts as an unbiased estimate of how the model performs on truly new data.

Hyperparameters vs Parameters **Parameters:** - Weights and biases (defining model)

Some Hyperparameters: - Learning rate in gradient descent - Number of epochs in perceptron update rule - Number of neurons in a neural network (we'll see what this means later) - Number of layers in a neural network (we'll see what this means later)

Summary

- **Training set** teaches the model.
- **Validation set** guides development decisions.
- **Test set** is the final report card.

  Caution: Protect Your Test Set

  The test set should be used **only once**, after all model decisions are finalized.

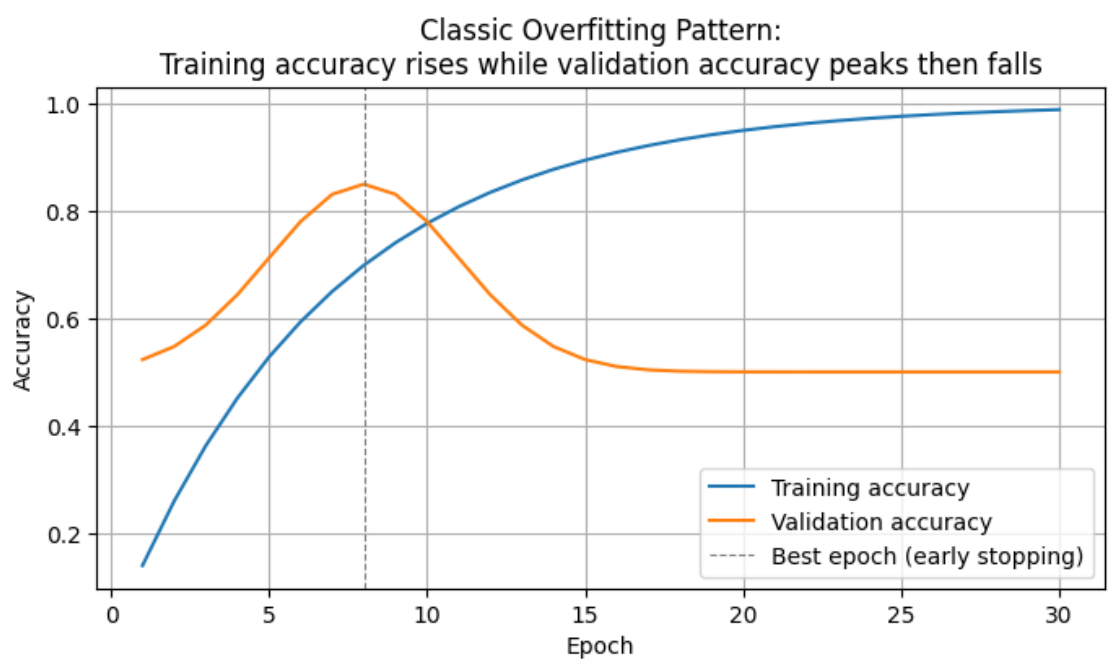
  Do **not** use the test set to:

  - choose model architectures
  - tune hyperparameters
  - decide when to stop training
  - compare multiple versions during development

If you repeatedly look at test performance while making decisions, the model indirectly adapts to the test data, and the reported results are no longer an unbiased measure of generalization.

Note: Using your test set to make model or training decisions is a form of *data leakage*. It can give a false sense of how well your model actually works.

[5]:
```
# Simulated overfitting curve - designed to show the classic pattern clearly
epochs = list(range(1, 31))

# Training accuracy: keeps climbing toward 1.0
train_acc = [1 - math.exp(-0.15 * e) for e in epochs]

# Validation accuracy: rises, peaks around epoch 8, then slowly falls
val_acc   = [0.5 + 0.35 * math.exp(-((e - 8)**2) / 18) for e in epochs]

plt.figure(figsize=(8, 4))
plt.plot(epochs, train_acc, label='Training accuracy')
plt.plot(epochs, val_acc,   label='Validation accuracy')
plt.axvline(x=8, color='gray', linestyle='--', linewidth=0.8, label='Best epoch␣
↪(early stopping)')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.title('Classic Overfitting Pattern:\nTraining accuracy rises while␣
↪validation accuracy peaks then falls')
plt.legend()
plt.grid(True)
plt.show()
```



Look at the plot above showing training and validation accuracy over epochs.

Q: At which epoch does validation accuracy reach its highest value?

`YOUR ANSWER HERE`

Q: After that epoch, training accuracy continues to rise but validation accuracy falls. What does this tell you about how well the model generalizes to new data?

`YOUR ANSWER HERE`

Q: If you could save only one version of the model (one epoch's weights), which epoch would you choose, and why?

`YOUR ANSWER HERE`

1.1.5 Overfitting

Overfitting occurs when a model learns the training data *too well* — including its noise and quirks — and fails to generalize to unseen data. A classic warning sign: near-zero training error alongside high validation or test error.

Overfitting often looks like:

- training metric keeps improving
- validation metric improves, then gets worse

We will simulate a plausible training/validation accuracy curve.

If training performance keeps improving but validation performance worsens, the model is overfitting. This is the opposite of generalation: a model **generalizes** when what it learned on training data extends reliably to new, unseen examples drawn from the same distribution.

Some Ways We Can Reduce Overfitting

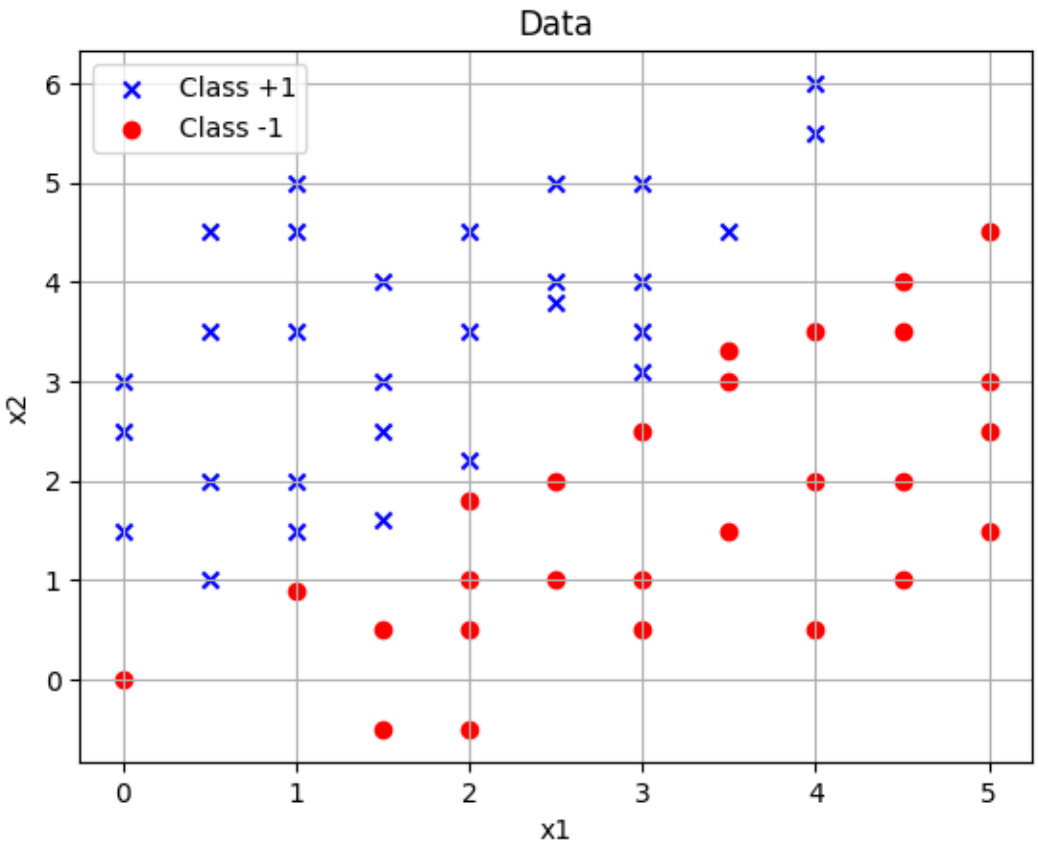
- **Use a validation set (early stopping)**
  Monitor validation loss during training and stop when it begins to worsen.
- **Simplify the model**
  Use fewer parameters (for example, fewer layers, fewer neurons, or a lower-degree model).
- **Add regularization**
  Add a small penalty to discourage very large weights, encouraging smoother solutions.
- **Increase the training data**
  More data makes it harder for the model to memorize noise.
- **Use data augmentation**
  Create additional training examples by slightly modifying existing ones (for example, flipping or rotating images).

Let's load an example to help us see these concepts concretely. This is an extension of the same toy data set from before.

[6]:
```python
# Toy perceptron example, with more data!
# Each input is [x1, x2], label is +1 if x2 > x1, else -1

X_toy_bigger = [
    # --- Original points ---
    [1.5, 4.0],
    [1.0, 2.0],
    [2.0, 1.0],
    [3.0, 5.0],
    [4.0, 2.0],
    [0.0, 0.0],
    [1.5, -0.5],

    # --- Class +1 (x2 > x1) ---
    [0.5, 2.0],
    [1.0, 3.5],
    [2.0, 4.5],
    [0.0, 1.5],
    [3.0, 4.0],
    [1.5, 3.0],
    [2.5, 5.0],
    [0.5, 1.0],
    [4.0, 5.5],
    [1.0, 1.5],
    [3.5, 4.5],
    [0.0, 2.5],
    [2.0, 3.5],
    [1.5, 2.5],
    [0.5, 3.5],
    [3.0, 3.5],
    [2.5, 4.0],
    [1.0, 4.5],
    [0.0, 3.0],
    [1.0, 5.0],
    [2.5, 3.8],
    [0.5, 4.5],
    [4.0, 6.0],

    # --- Class -1 (x2 <= x1) ---
    [3.0, 1.0],
    [4.5, 2.0],
    [2.0, 0.5],
    [5.0, 3.0],
    [3.5, 1.5],
    [4.0, 0.5],
    [2.5, 1.0],
    [5.0, 4.5],
```

```
    [4.5, 3.5],
    [3.0, 2.5],
    [1.5, 0.5],
    [2.5, 2.0],
    [4.0, 3.5],
    [3.5, 3.0],
    [5.0, 1.5],
    [5.0, 2.5],
    [4.5, 1.0],
    [3.0, 0.5],
    [2.0, -0.5],
    [4.5, 4.0],

    # --- Noisy points near x2  x1 ---
    [2.0, 2.2],
    [3.0, 3.1],
    [1.5, 1.6],
    [2.0, 1.8],
    [3.5, 3.3],
    [1.0, 0.9],
]

y_toy_bigger = [
    # Original
    1, 1, -1, 1, -1, -1, -1,
    # Class +1
    1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
    # Class -1
    -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,␣
↪-1,
    # Noisy
    1, 1, 1, -1, -1, -1,
]

# Separate points by class for coloring
def plot_points(X_points, y_labels):
    X_pos = [x for x, y in zip(X_points, y_labels) if y == 1]
    X_neg = [x for x, y in zip(X_points, y_labels) if y == -1]

    plt.scatter([x[0] for x in X_pos], [x[1] for x in X_pos], color='blue',␣
↪marker='x', label='Class +1')
    plt.scatter([x[0] for x in X_neg], [x[1] for x in X_neg], color='red',␣
↪marker='o',  label='Class -1')

    plt.xlabel('x1')
    plt.ylabel('x2')
    plt.title('Data')
```

```
    plt.legend()
    plt.grid(True)
    plt.show()

plot_points(X_toy_bigger, y_toy_bigger)
```



1.1.6 Training Curve vs. Learning Curve

Training Curve — plots error or other metric on the *training set* as a function of steps / epochs or training set size. - A very low training error sounds good, but alone it tells us nothing about generalization.

Learning Curve — plots *both* training and test (or validation) error as training set size grows. - As training size increases, we expect: - Training error to **rise slightly** (harder to perfectly fit more data) - Test error to **fall** (model sees more of the true distribution) - When the two curves **converge at a low error**, the model is generalizing well. - When training error is low but test error stays high, the model is **overfitting**.

Before we explore how the error rate of the perceptron changes throughout the training and validation, we need to consider how to split the data set.

Implementing train/test split

[7]:
```
# Use type hints for function parameter and return type
from typing import List, Sequence, Tuple

def train_test_split_v1(
    X: Sequence[Sequence[float]],
    y: Sequence[int],
    test_ratio: float = 0.2,
    seed: int = 0,
    shuffle: bool = True,
) -> Tuple[List[List[float]], List[List[float]], List[int], List[int]]:
    '''
    Split a dataset into training and test subsets.

    Optionally shuffles the data before splitting so that the test set is
    a random sample rather than always the last `test_ratio` fraction of
    the original order.

    X: list of input features
    y: list of target labels
    test_ratio: fraction of the data to put in the test set. Must be between 0␣
↪and 1. Default is 0.2 (20%).
    seed: Random seed used for shuffling, so results are reproducible. Default␣
↪is 0.
    shuffle: If True, randomly shuffle the data before splitting.
            If False, take the first (1 - test_ratio) fraction for training and␣
↪the rest for testing. Default is True.
    '''
    indices = list(range(len(y)))

    if shuffle:
        # Create a local instance random number generator to randomly shuffle
        rng = random.Random(seed)
        rng.shuffle(indices)

    n_test = max(1, round(len(y) * test_ratio))
    test_indices  = indices[:n_test]
    train_indices = indices[n_test:]

    X_list = [list(row) for row in X]
    y_list = [int(v)    for v in y]

    return (
        [X_list[i] for i in train_indices],
        [X_list[i] for i in test_indices],
        [y_list[i] for i in train_indices],
        [y_list[i] for i in test_indices],
```

```
    )
```

[8]:
```
X_train, X_test, y_train, y_test = train_test_split_v1(X_toy_bigger,␣
↪y_toy_bigger, test_ratio=0.2, seed=42)

def class_counts(y, split_name):
    '''
    Count the number of positive and negative labeled data points in a data set

    y: list of labels
    split_name: descriptive string for which list of labels is being used
    '''
    pos = y.count(1)
    neg = y.count(-1)
    total = len(y)
    print(f"{split_name} ({total} points):")
    print(f"  Class +1: {pos}  ({100*pos/total:.1f}%)")
    print(f"  Class -1: {neg}  ({100*neg/total:.1f}%)")

class_counts(y_toy_bigger,   "Full dataset")
class_counts(y_train, "Train set   ")
class_counts(y_test,  "Test set    ")
```

```
Full dataset (56 points):
  Class +1: 29  (51.8%)
  Class -1: 27  (48.2%)
Train set    (45 points):
  Class +1: 21  (46.7%)
  Class -1: 24  (53.3%)
Test set     (11 points):
  Class +1: 8  (72.7%)
  Class -1: 3  (27.3%)
```

How much does class balance vary across random splits?

[9]:
```
# How much does class balance vary across random splits (no stratification)?
print("Seed | Test +1 | Test -1 | Train +1 | Train -1")
print("-" * 50)
for seed in range(20):
    X_tr, X_te, y_tr, y_te = train_test_split_v1(X_toy_bigger, y_toy_bigger,␣
↪test_ratio=0.2, seed=seed)
    print(
        f"  {seed:2d} |"
        f"   {y_te.count(1):3d}   |"
        f"   {y_te.count(-1):3d}   |"
        f"    {y_tr.count(1):3d}   |"
        f"    {y_tr.count(-1):3d}"
    )
```

```
print()
print(f"Full dataset: +1={y_toy_bigger.count(1)}, -1={y_toy_bigger.count(-1)}")
print(f"Ideal test set (20%): +1 {round(y_toy_bigger.count(1)*0.2)},␣
↪-1 {round(y_toy_bigger.count(-1)*0.2)}")
```

```
Seed | Test +1 | Test -1 | Train +1 | Train -1
--------------------------------------------------
   0 |     5   |     6   |     24   |     21
   1 |     4   |     7   |     25   |     20
   2 |     8   |     3   |     21   |     24
   3 |     9   |     2   |     20   |     25
   4 |     7   |     4   |     22   |     23
   5 |     4   |     7   |     25   |     20
   6 |     4   |     7   |     25   |     20
   7 |     6   |     5   |     23   |     22
   8 |     7   |     4   |     22   |     23
   9 |     5   |     6   |     24   |     21
  10 |     3   |     8   |     26   |     19
  11 |     5   |     6   |     24   |     21
  12 |     5   |     6   |     24   |     21
  13 |     5   |     6   |     24   |     21
  14 |     6   |     5   |     23   |     22
  15 |     6   |     5   |     23   |     22
  16 |     7   |     4   |     22   |     23
  17 |     1   |    10   |     28   |     17
  18 |     8   |     3   |     21   |     24
  19 |     3   |     8   |     26   |     19

Full dataset: +1=29, -1=27
Ideal test set (20%): +1 6, -1 5
```

Which seed do we trust? We shouldn't have to choose — that's what stratification fixes.

What is Stratification? **Stratification** ensures that each split (train, test, or validation) contains roughly the **same proportion of each class** as the full dataset.

For example, if the full dataset is 60% Class +1 and 40% Class -1, a stratified split guarantees your train and test sets are also approximately 60/40 — regardless of the random seed used.

Without stratification, a random shuffle might accidentally concentrate one class in the test set, making evaluation misleading.

[10]:
```
def train_test_split_v2(
    X: Sequence[Sequence[float]],
    y: Sequence[int],
    test_ratio: float = 0.2,
    seed: int = 0,
    shuffle: bool = True,
```

```
) -> Tuple[List[List[float]], List[List[float]], List[int], List[int]]:
    '''
    Split data set with stratification: preserves class proportions in each␣
↪split.

    X: list of input features
    y: list of target labels
    test_ratio: fraction of the data to put in the test set. Must be between 0␣
↪and 1. Default is 0.2 (20%).
    seed: Random seed used for shuffling, so results are reproducible. Default␣
↪is 0.
    shuffle: If True, randomly shuffle the data before splitting.
             If False, take the first (1 - test_ratio) fraction for training and␣
↪the rest for testing. Default is True.
    '''
    # Create a local instance random number generator, in case shuffle is True.
    rng = random.Random(seed)
    # Group indices by class label
    labels = list(set(y))
    class_indices = {label: [i for i, v in enumerate(y) if v == label] for␣
↪label in labels}

    train_indices = []
    test_indices  = []

    for label in labels:
        idxs = class_indices[label][:]
        if shuffle:
            rng.shuffle(idxs)
        n_test = max(1, round(len(idxs) * test_ratio))
        test_indices  += idxs[:n_test]
        train_indices += idxs[n_test:]

    if shuffle:
        rng.shuffle(train_indices)
        rng.shuffle(test_indices)

    X_list = [list(row) for row in X]
    y_list = [int(v)    for v in y]

    return (
        [X_list[i] for i in train_indices],
        [X_list[i] for i in test_indices],
        [y_list[i] for i in train_indices],
        [y_list[i] for i in test_indices],
    )
```

[11]:
```
print("       -v1 (no stratify)-    -v2 (stratified)-")
print("Seed | Test +1 | Test -1 | Test +1 | Test -1")
print("-" * 55)
for seed in range(20):
    _, _, _, y_te_v1 = train_test_split_v1(X_toy_bigger, y_toy_bigger,␣
 ↪test_ratio=0.2, seed=seed)
    _, _, _, y_te_v2 = train_test_split_v2(X_toy_bigger, y_toy_bigger,␣
 ↪test_ratio=0.2, seed=seed)
    print(
        f"  {seed:2d} |"
        f"   {y_te_v1.count(1):3d}   |"
        f"   {y_te_v1.count(-1):3d}   |"
        f"   {y_te_v2.count(1):3d}   |"
        f"   {y_te_v2.count(-1):3d}"
    )

print()
print(f"Full dataset: +1={y_toy_bigger.count(1)}, -1={y_toy_bigger.count(-1)}")
print(f"Ideal test set (20%): +1 {round(y_toy_bigger.count(1)*0.2)},␣
 ↪-1 {round(y_toy_bigger.count(-1)*0.2)}")
```

```
       -v1 (no stratify)-    -v2 (stratified)-
Seed | Test +1 | Test -1 | Test +1 | Test -1
-------------------------------------------------------
   0 |     5   |     6   |     6   |     5
   1 |     4   |     7   |     6   |     5
   2 |     8   |     3   |     6   |     5
   3 |     9   |     2   |     6   |     5
   4 |     7   |     4   |     6   |     5
   5 |     4   |     7   |     6   |     5
   6 |     4   |     7   |     6   |     5
   7 |     6   |     5   |     6   |     5
   8 |     7   |     4   |     6   |     5
   9 |     5   |     6   |     6   |     5
  10 |     3   |     8   |     6   |     5
  11 |     5   |     6   |     6   |     5
  12 |     5   |     6   |     6   |     5
  13 |     5   |     6   |     6   |     5
  14 |     6   |     5   |     6   |     5
  15 |     6   |     5   |     6   |     5
  16 |     7   |     4   |     6   |     5
  17 |     1   |    10   |     6   |     5
  18 |     8   |     3   |     6   |     5
  19 |     3   |     8   |     6   |     5

Full dataset: +1=29, -1=27
Ideal test set (20%): +1 6, -1 5
```

Now that we have a reliable split, we'll use the bigger toy dataset to explore three questions about how a model learns:

1. **Training curve** — how does training and test error change epoch by epoch? *(Answer: training error falls quickly; the gap between train and test error shows how well the model generalizes to new data.)*
2. **Learning curve** — how does test error change as we increase the training set size? *(Answer: more data gives lower test error and smaller gap between train and test.)*
3. **Decision boundary comparison** — what does the *shape* of the learned boundary look like for a tiny vs. large training set? *(Answer: with little data, the boundary may fit the few points it saw but miss the true underlying pattern.)*

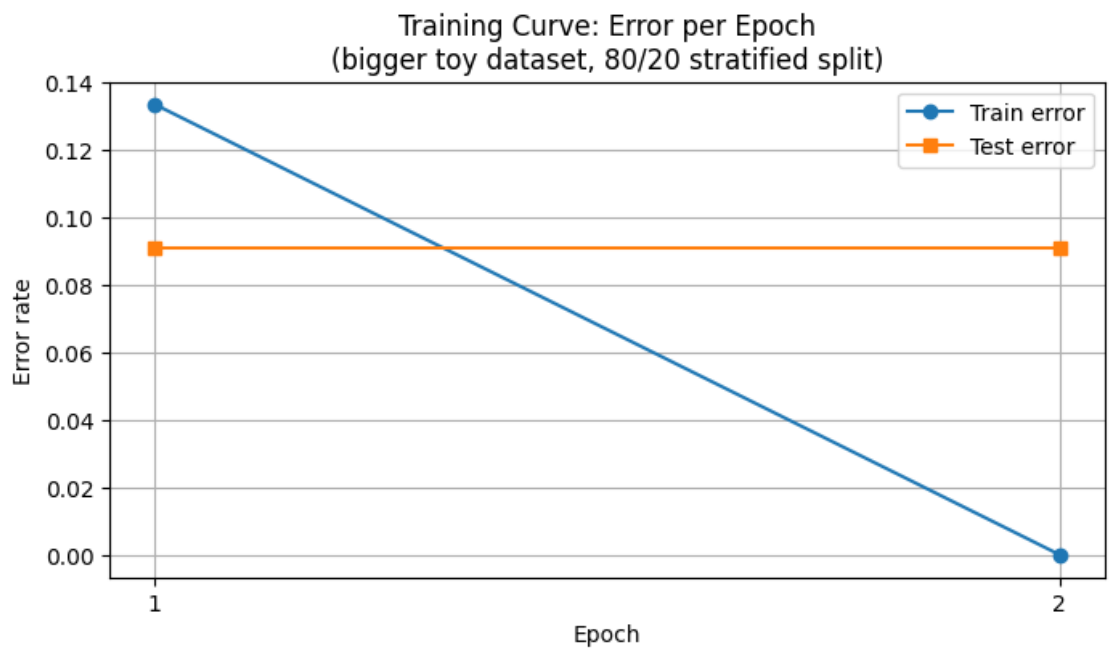
We'll train the perceptron on two different versions of the same dataset: - A **tiny training set** (only 15% of the data) — the model has very little to learn from. - A **larger training set** (80% of the data) — the model has much more information.

[12]:
```
def perceptron_train_eval_with_history(X, y, num_epochs=10,X_test=None,␣
↪y_test=None):
    '''
    Train a perceptron and record loss, accuracy, precision, and recall
    after every individual data point is processed (across all epochs).

    X: list of [x1, x2] feature pairs
    y: list of labels in {-1, +1}
    num_epochs: number of full passes over the data
    X_test, y_test: optional test sets for calculating test accuracty

    Returns
    w, b : final weights and bias
    history : dictionary with keys 'loss', 'accuracy', 'precision', 'recall',
              each a list of length (num_epochs * len(X)). Also has a key␣
↪'test_accuracy' (one value per epoch) if X_test is provided.
    '''
    w = [0.0, 0.0]
    b = 0.0

    history = {'loss': [], 'accuracy': [], 'precision': [], 'recall': [],␣
↪'test_accuracy': []}

    rng = random.Random(42)

    for epoch in range(num_epochs):
        # Shuffle a new order each epoch
        combined = list(zip(X, y))
        rng.shuffle(combined)

        for x, y_true in combined:
```

```
            # Calculate perceptron score and apply update rule
            score  = w[0] * x[0] + w[1] * x[1] + b
            y_pred = 1 if score >= 0 else -1

            if y_pred != y_true:
                w[0] += y_true * x[0]
                w[1] += y_true * x[1]
                b    += y_true

            # Calculate current model predictions for all training data
            all_preds = perceptron_predict_all(X, w, b)

            # Total perceptron loss = sum of max(0, -y*z) across all points
            total_loss = sum(
                max(0.0, -yt * (w[0]*xt[0] + w[1]*xt[1] + b))
                for xt, yt in zip(X, y)
            )

            acc, prec, rec = compute_metrics(y, all_preds)

            history['loss'].append(total_loss)
            history['accuracy'].append(acc)
            history['precision'].append(prec)
            history['recall'].append(rec)

        # Record test accuracy once per epoch
        if X_test is not None and y_test is not None:
            test_preds = perceptron_predict_all(X_test, w, b)
            test_acc, _, _ = compute_metrics(y_test, test_preds)
            history['test_accuracy'].append(test_acc)

        if history['accuracy'][-1] == 1.0: break

    return w, b, history
```

[13]:
```
# Training curve: how model error changes as training progresses (epoch by␣
↪epoch)
X_train, X_test, y_train, y_test = train_test_split_v2(
    X_toy_bigger, y_toy_bigger, test_ratio=0.2, seed=42
)

w, b, history = perceptron_train_eval_with_history(
    X_train, y_train, num_epochs=20,
    X_test=X_test, y_test=y_test
)

n_pts = len(X_train)
```

```
num_epochs_run = len(history['test_accuracy'])

# Sample epoch-end values from the per-point history
epoch_train_err = [1.0 - history['accuracy'][n_pts*(e+1)-1] for e in␣
 ↪range(num_epochs_run)]
epoch_test_err  = [1.0 - a for a in history['test_accuracy']]

epochs = list(range(1, num_epochs_run + 1))
plt.figure(figsize=(8, 4))
plt.plot(epochs, epoch_train_err, marker='o', label='Train error')
plt.plot(epochs, epoch_test_err,  marker='s', label='Test error')
plt.xlabel('Epoch')
plt.xticks([1, 2])
plt.ylabel('Error rate')
plt.title('Training Curve: Error per Epoch\n(bigger toy dataset, 80/20␣
 ↪stratified split)')
plt.legend()
plt.grid(True)
plt.show()
```



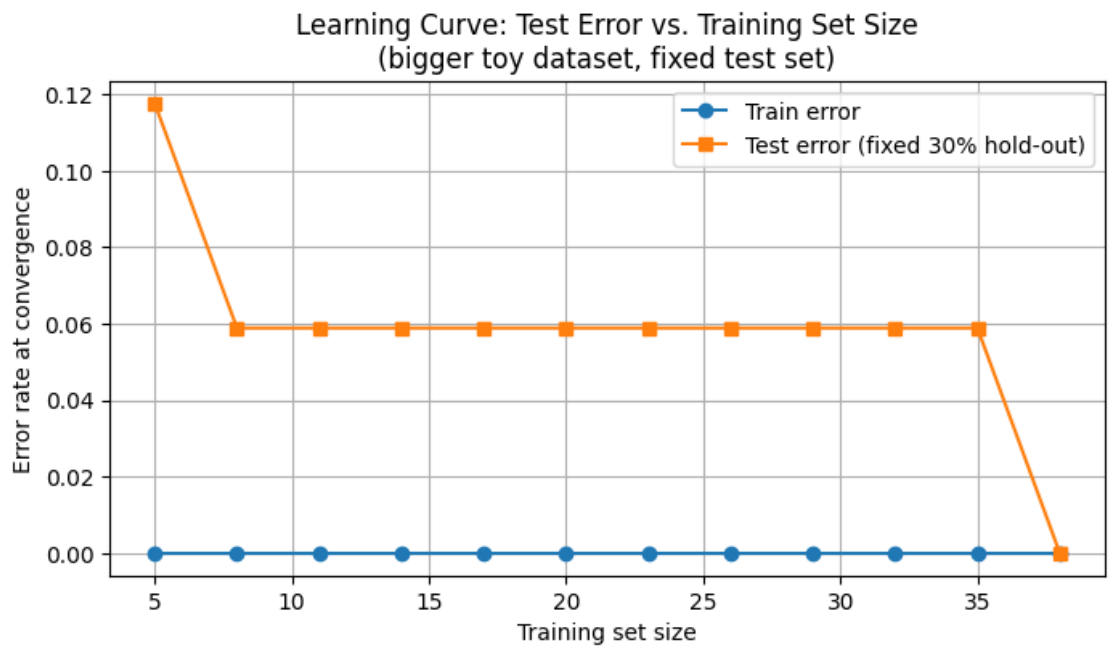
Some observations about the curves:

- **Early stopping of training**
  - At epoch 1 the model has completed one full pass over the training data but hasn't yet classified all points correctly. It makes about 6 errors on ~44 training points (13.6%).
  - The perceptron reaches zero training error by the end of epoch 2 and stops.

  - The x-axis only goes to 2 even though `num_epochs=20` was requested.
- **Test error is flat**
  - Test error is the same at epoch 1 and at epoch 2, despite train error decreasing.
  - The test set has ~11 points (20% of ~55). At epoch 1, the boundary already correctly classifies 10 of those 11 test points. At epoch 2, even though the boundary shifts enough to fix all remaining training errors, the one misclassified test point stays on the wrong side. The boundary adjustment helped training but did not change how the test point was classified.
  - Fitting the training data perfectly does not guarantee better generalization. The model has converged to a valid separating hyperplane for the training data but not the one that happens to classify this particular test set perfectly.
  - The 9% persistent test error tells us something real: the model has learned a boundary that generalizes imperfectly, and no amount of additional training epochs will fix it. The only remedy is more *training data* or a different hyperparameter (e.g. initialization of random seed).
- **How can train error be higher than test error?**
  - Normally we expect train error to be less than or equalt to test error, since the model was trained on those points, so it should "know" them better.
  - The 20-point test set contains 11 relatively easy-to-classify points (they may lie far from the true boundary). On the other hand, the 44 training points include some near the boundary that the model keeps getting wrong during the first epoch. By the end of epoch 1 the model has mostly converged but a few boundary-adjacent training points are still misclassified — while all test points happen to already be on the right side of the partially-trained boundary.
  - Low test error early on does not mean the model is "good". Rather, it means this particular train/test split produced a test set that was easier than average. This is exactly the kind of accident that **stratification helps prevent** for class balance, but cannot prevent for "easy vs. hard" cases.

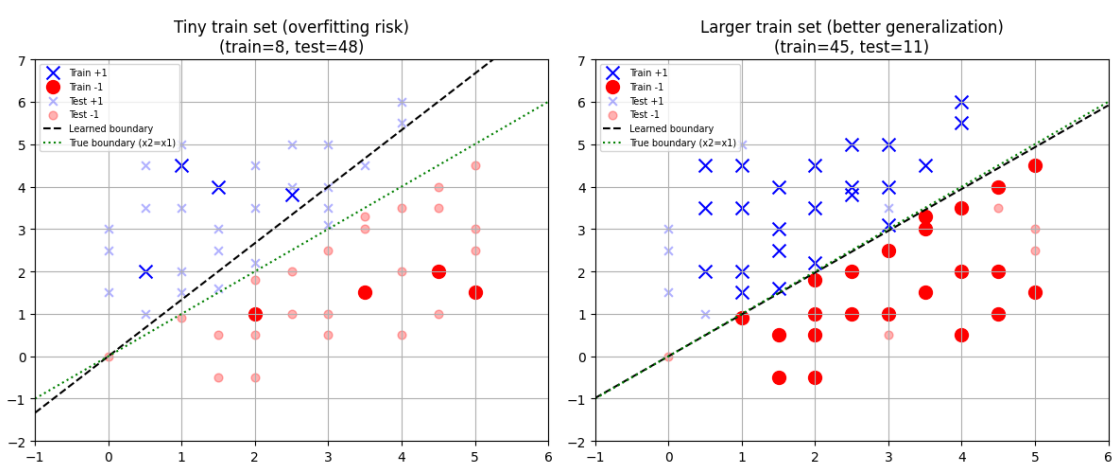
[14]:
```
# Learning curve: how test error changes as we give the model more training data
# Fixed test set, grow only the training subset

# Reserve a fixed 30% as the test set - this never changes.
X_train_pool, X_test_fixed, y_train_pool, y_test_fixed = train_test_split_v2(
    X_toy_bigger, y_toy_bigger, test_ratio=0.30, seed=42
)

# Sweep over growing subsets of the training pool.
# Use train_test_split_v2 for each size so that class balance is preserved
# at every training set size (stratified subsampling).
min_train = 5
max_train = len(X_train_pool)
step      = 3

all_sizes  = list(range(min_train, max_train + 1, step))
train_errs = []
test_errs  = []
```

```
for n_train in all_sizes:
    if n_train >= len(X_train_pool):
        # Use the entire training pool
        X_tr, y_tr = X_train_pool, y_train_pool
    else:
        # Stratified subsample: take n_train points from the pool,
        # preserving the positive/negative class ratio.
        discard_ratio = 1.0 - n_train / len(X_train_pool)
        X_tr, _, y_tr, _ = train_test_split_v2(
            X_train_pool, y_train_pool,
            test_ratio=discard_ratio,
            seed=42
        )

    w_lc, b_lc, hist_lc = perceptron_train_with_history(X_tr, y_tr,␣
 ↪num_epochs=50)

    # Training error at convergence
    train_err = 1.0 - hist_lc['accuracy'][-1]

    # Test error on the FIXED test set (same 30% every time)
    test_preds = perceptron_predict_all(X_test_fixed, w_lc, b_lc)
    test_acc, _, _ = compute_metrics(y_test_fixed, test_preds)
    test_err = 1.0 - test_acc

    train_errs.append(train_err)
    test_errs.append(test_err)

plt.figure(figsize=(8, 4))
plt.plot(all_sizes, train_errs, marker='o', label='Train error')
plt.plot(all_sizes, test_errs,  marker='s', label='Test error (fixed 30%␣
 ↪hold-out)')
plt.xlabel('Training set size')
plt.ylabel('Error rate at convergence')
plt.title('Learning Curve: Test Error vs. Training Set Size\n(bigger toy␣
 ↪dataset, fixed test set)')
plt.legend()
plt.grid(True)
plt.show()
```



Some observations about the curves:

- **Very low training error**
  - Training error stays near 0 for all training set sizes
  - The perceptron always converges to zero train error on linearly separable data.
- **Generalization improves with larger training set size**
  - Test error starts high when the training set is tiny
  - Few training points likely means poor generalization
  - Test error monotonically decreases as more training data is provided
  - The gap between the train error and test error curves is the "generalization gap". It is large when the training set is small, shrinking toward zero as training size grows.

[15]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

for ax, ratio, title in zip(
    axes,
    [0.85, 0.20],
    ["Tiny train set (overfitting risk)", "Larger train set (better␣
↪generalization)"]
):
    X_tr, X_te, y_tr, y_te = train_test_split_v2(X_toy_bigger, y_toy_bigger,␣
↪test_ratio=ratio, seed=42)
    w, b, _ = perceptron_train_with_history(X_tr, y_tr, num_epochs=20)

    # Plot all points, highlighting which were in training
    X_pos_tr = [x for x, y in zip(X_tr, y_tr) if y ==  1]
```

```
    X_neg_tr = [x for x, y in zip(X_tr, y_tr) if y == -1]
    X_pos_te = [x for x, y in zip(X_te, y_te) if y ==  1]
    X_neg_te = [x for x, y in zip(X_te, y_te) if y == -1]

    ax.scatter([x[0] for x in X_pos_tr], [x[1] for x in X_pos_tr],␣
↪color='blue',  marker='x', s=100, label='Train +1')
    ax.scatter([x[0] for x in X_neg_tr], [x[1] for x in X_neg_tr], color='red',␣
↪  marker='o', s=100, label='Train -1')
    ax.scatter([x[0] for x in X_pos_te], [x[1] for x in X_pos_te],␣
↪color='blue',  marker='x', s=40, alpha=0.3, label='Test +1')
    ax.scatter([x[0] for x in X_neg_te], [x[1] for x in X_neg_te], color='red',␣
↪  marker='o', s=40, alpha=0.3, label='Test -1')

    import numpy as np
    x1_vals = np.linspace(-1, 6, 100)
    if w[1] != 0:
        x2_vals = -(w[0] * x1_vals + b) / w[1]
        ax.plot(x1_vals, x2_vals, 'k--', label='Learned boundary')

    # True boundary for reference
    ax.plot(x1_vals, x1_vals, 'g:', label='True boundary (x2=x1)')
    ax.set_xlim(-1, 6)
    ax.set_ylim(-2, 7)
    ax.set_title(f"{title}\n(train={len(X_tr)}, test={len(X_te)})")
    ax.legend(fontsize=7)
    ax.grid(True)

plt.tight_layout()
plt.show()
```